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Lab 07: Reverse Polish Notation

What was the Lab about?

We created both a scanner and linked list stack which we then used to turn infix expressions into postfix expressions and evaluate them. We then expanded on the steps before by allowing expressions to be entered into the console and having the console return the evaluated expression.

Part 1:

The scanner takes input and returns it as a String array of tokens.

Part 2:

The linked list stack allows for the creation of a stack of elements, providing the user with methods top(), isEmpty(), and bottom() to check on the stack and the methods push() and pop() to manipulate it.

Part 3:

Each token in the array, generated using the input, is checked individually through the multiple steps from the given infix-to-postfix algorithm and is either added to a result String or pushed into a stack to be added to the result String at a later point. At the end, the result String is returned. Unfortunately, we ran into some issues in instances where parentheses were included in the equation. In these cases an open parenthesis was added to the result String and the order after said parenthesis was incorrect.

Part 4:

We take a given postfix expression and utilize the algorithm provided and evaluate it. The expression is looped through, pushing operands into a stack until an operator is reached. At this point the top two operands of the stack are popped and the operator is used on them. The result of this is then pushed onto the stack again to be used as an operand again, once another operator is reached.

Part 5

We take input from the console and run it through the code we used in Part 4 and return the output directly to the console.